

IYAD EL KHOURY

Status: Incoming Master Student at l'ÉNS de Lyon

Fields: Discrete Math, Probability Theory, Machine Learning, Algorithmics

Programming Languages: Python, C++, Coq, Latex

Languages: English C2, French C2, Dutch B2, Arabic B1

Hobbies: Basketball, board & card games, cooking

Palaiseau, France

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Born: 05/12/2005

[My personal website](#) & [LinkedIn](#)

Education

Exchange Studies - ETH Zürich

Fall Semester 2025

- Studied at ETH Zürich as an exchange student in the Department of Mathematics (D-MATH).
- Relevant Courses:** Machine Learning for Finance, Algebraic Methods in Combinatorics, Measure Theory, Commutative Algebra, Algebraic Number Theory.

Bachelor Student - École Polytechnique, Palaiseau, France

2023 - 2026

- CGPA : 4.11/4.3**, ranked **11th out of 155** overall & **5th out of 70** in my major Mathematics & Computer Science.
- Relevant Courses:**
 - Analysis & Probability:** Real Analysis, Topology, Series of functions & Fourier Series, Differential Geometry, Numerical Analysis, Probability Theory & Statistics, Stochastic Processes,
 - Computer Science:** Design & Analysis of Algo., Complexity, Formal Languages & Methods, Programming in Python I & II, Object-Oriented Programming in C/C++, Logic & Proofs, Computer Arch., Machine Learning
 - Algebra & Combinatorics:** Linear Algebra I & II, Euclidean Spaces & Quad. Forms, Algo. for discrete math
- Math committee (BMX):** I organized bimonthly popular seminars, presented by both researchers and students, on various mathematical topics. Some previous themes : Paradoxes in Math, Cryptography, p-adic numbers ...

Awards

ÉNS Lyon entrance examinations - ÉNS Lyon

May - July 2026

- ÉNS Lyon is a prestigious research oriented school in France, whose selection procedure is done via competitive entrance examinations. I was admitted in the program ranking **2nd** overall in the scientific disciplines.
- Being admitted through such exams provides me with a 4 year financial support from the french government.

SEMP scholarship - Zürich, Switzerland

Fall Semester 2025

- Received an Erasmus scholarship as a financial support for my exchange semester at ETH Zürich.

Prosperity 3 challenge - IMC

April 2025

- The prosperity challenge organized by IMC is a Algorithmic Trading Challenge aimed to introduce students to the strategies and mathematics used in modern trading.
- Achieved a global ranking of **29th out of 12620** and **2nd** in France with my team *Musashi's Island*. We solved both theoretical game theory problems and algorithmic trading problems which consisted of finding optimal strategies to trade certain asset classes given historical data.
- Prosperity 3 allowed me to apply my passion for mathematical research in an exciting and competitive setting.

Research Projects & Internships

Bachelor Thesis, under Prof. Anurag Bishnoi - TU Delft, Discrete Math & Optimization

Jan. 2026- March 2026

- My thesis was focused on a combinatorial sequence in game theory, called the *Hat Guessing Number*, the initial goal was to take a novel geometric approach to the problem.
- By studying polarities of certain finite projective spaces, I constructed a geometric strategy to the Hat guessing game. I provided a new proof of an existing theorem which further generalizes to a larger class of graphs.

Polymath Summer Camp, by Prof. Guillaume Dubach - École Polytechnique

June 2025

- The program focused on random matrix theory and applications to ML, random permutations & combinatorics.
- The program consisted of guest lectures, group discussions and a final presentation on a theorem of choice to the other students. I presented the probabilistic approach to proving the hook length formula.

GNNs in combinatorics - École Polytechnique

March-June 2025

- I designed and implemented a GNN to approximate solutions to the so called *burning problem*, providing a new perspective on the problem. Presented the project to prof. Johannes Lutzeyer at LIX.

Research internship, under Prof. Lorenzo Fantini - Polytechnique math dep. (CMLS)

January-June 2025

- Under the supervision of Prof. Lorenzo Fantini, I studied Ramsey Numbers, a combinatorial sequence at the intersection of Discrete Mathematics, Optimization, Computer Science and Probability.
- We initially explored classical probabilistic methods, which have many beautiful applications in combinatorics, before turning to more subtle geometric arguments using the theory of incidence geometry.
- I wrote a final rapport and gave an oral defense to discuss the topic with researchers from CMLS.

Research internship, under Prof. Jeremie Bettinelli - Polytechnique CS dep. (LIX)

June-August 2024

- Researched the "Burning number Conjecture", an open problem in graph theory and optimization. The internship was a group work in addition to weekly meetings with our supervisor to discuss approaches to a solution.